MASTER'S PHYSICS THESIS

Evolution of Quantum Open System under Continuous Measurement

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« La gloire de Dieu, c'est de cacher les choses; La gloire des rois, c'est de sonder les choses. »
— **Proverbes 25:2**

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I Theory of open quantum systems and continuous measurements

I.i) Introduction and motivations

The paper you read will read is purposely non-exhaustive ; it consists of a pedagogical and demonstrative introduction to *quantum continuous measurements*. Talented searchers made thorough and detailed papers on the subjects we will tackle, but we wanted to allow mere students or uninitiated people to understand the basics of weak measurements.

Quantum mechanics is the most fundamental physical theory humankind has developed due to accurate description at the smallest scale of the Universe. The primary results of quantum physics are obtained while considering a system isolated in the void from any perturbation or interaction. However, it seems trivial that any real physical system is "feeling" its surroundings (light rays or magnetic field hitting the system, coupling with a cavity, etc...) thus we ought to describe the impact of the environment on a system ; the couple System-Environment is called as *Open* Quantum System (OQS). Before diving more into this theory we should describe what we call a measurement [1, 2], the postulate of quantum measurement states that we can choose any basis to represent a system, living mathematically in an Hilbert space, $|\psi\rangle \in \mathcal{H}$:

$$|\psi\rangle = \sum_n c_n |\phi_n\rangle \quad ; \; c_n \in \mathbb{C}$$

We can look at which eigen-state $|\phi_n\rangle$ the system is in, doing so will (always) put the system in one of these states and we know the probability to get this state: $|c_n|^2$. This kind of measurement is called a *von Neumann measurement* where the result is a state among the set of eigen-states. Formally, it is a Projection-Valued Measurement (PVM) where the projector associated with the probability to find the result n is $\mathbb{P}_n = \langle\psi| P_n |\psi\rangle$. More general *quantum measurement* or Positive-Operator Valued Measures (POVM) Π_n can be defined by $\mathbb{P}_n = \mathrm{Tr}\left[\rho\Pi_n\right]$ where we use a generalization of quantum states into density matrix ρ, which for pure states is $\rho_{pure} = |\psi\rangle\langle\psi|$. This new formalism of density matrix has importance revealed when dealing with OQS, please find Appendix A for further understanding.

I.ii) Open quantum systems basics

I.iii) Obtaining Stochastic Master Equations (SME)

A Quantum Measurement Theory

The introduction about measurement theory in I.i) and this appendix have been constructed from the work of *K.Jacobs* [1] and *H.M.Wiseman* [2].

References

[1] Kurt Jacobs. *Quantum Measurement Theory and its Applications*. en. Cambridge, England: Cambridge University Press, Aug. 2014.

[2] Howard Mark Wiseman. "Quantum trajectories and feedback". PhD thesis. University of Queensland Library. DOI: 10.14264/366355. URL: http://dx.doi.org/10.14264/366355.